Lineage-Aware Hierarchical Deep Learning for Imbalanced Multi-Class Classification of

Peripheral Blood Cells in Hematological Diagnostics

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Dissertation Proposal

Background

Examining a peripheral blood smear under the microscope is a core step in diagnosing hematological diseases, including leukemia subtypes. A trained examiner inspects each nucleated cell and assigns it to a morphological category based on nucleus shape, chromatin pattern, cytoplasm, and granularity. This differential count is informative, but it is repetitive, time consuming, and subject to variation between observers, particularly for rare or immature cells whose appearance overlaps with neighbouring categories.

Convolutional neural networks have shown strong results in this setting because they learn discriminative features directly from pixels, and deep models can already match expert recognition of blast cells in acute myeloid leukaemia [2, 3]. Most published systems, however, treat the task as flat classification: every cell type is an independent label, and the biological relationship between types is ignored. This is a missed opportunity, because blood cells are not unordered categories. They sit on well defined haematopoietic lineages, and immature myeloid cells in particular form an ordered maturation continuum [6].

This dissertation proposes a deep learning approach deliberately shaped by this biology. Instead of a single flat classifier, it develops a *lineage-aware hierarchical model* that first separates cells by haematopoietic lineage and then resolves the fine cell type within that lineage. Because the chosen image dataset is severely imbalanced, the approach is paired with an imbalance-robust training objective, and predictions are made interpretable using gradient-based saliency maps so a clinician can see which morphological regions drove each decision.

Aim and Objectives

The aim is to develop and evaluate a lineage-aware hierarchical deep learning model for automated multi-class classification of single peripheral blood cell images, robust to severe class imbalance and supported by visual explanations. The objectives are:

- Review the literature on single-cell blood image classification, hierarchical classification, class-imbalance handling, and explainable AI.
- Analyse the MLL23 peripheral blood image dataset and characterise its 18 classes and imbalance profile.
- Preprocess the single-cell images and define a two-level label hierarchy that reflects haematopoietic lineage.
- Develop a transfer-learning backbone with two classification heads, one for the lineage super-class and one for the fine cell type.
- Integrate a class-balanced loss so rare cell types are not ignored during training.
- Evaluate using accuracy, macro precision, recall, F1-score, balanced accuracy, and the confusion matrix, comparing against a flat baseline classifier.
- Generate Grad-CAM saliency maps and confirm that predictions rely on morphologically meaningful regions.

Research Question

Does a lineage-aware hierarchical classifier, combined with an imbalance-robust training objective, improve fine-grained classification of single peripheral blood cell images, especially for rare cell types, compared with a flat transfer-learning classifier on the same data?

Literature Review

Single-cell blood image classification is an active area within biomedical computer vision. Early systems relied on hand-crafted morphological features, but convolutional neural networks now dominate because they learn the relevant visual cues automatically [3]. Recent architectures continue to push accuracy on public smear datasets [4], yet a recurring finding is that performance is limited less by network architecture than by the quantity, quality, and balance of annotated single-cell data [5]. The dataset used here directly addresses this bottleneck: it is the largest publicly available expert-annotated peripheral blood single-cell collection, and its authors note that rare cell types occur infrequently in real blood, producing a strongly imbalanced distribution that reflects clinical reality but is difficult for standard classifiers [1].

Two methodological themes are central to this proposal. First, hierarchical classification exploits known relationships between labels by predicting coarse categories before fine ones, which shares representations usefully and reduces biologically implausible errors; hierarchical models have been shown to outperform flat classifiers on medical images, especially when data are limited [7, 8]. Second, class-imbalance methods such as the class-balanced loss based on the effective number of samples [10], focal loss [11], and more recent margin- and triplet-based objectives for imbalanced medical classification [9] are designed so minority classes are not overwhelmed during optimisation. Because the application is clinical, explainability also matters: gradient-based methods such as Grad-CAM highlight the regions responsible for a prediction [13], and have recently been used to validate blood-cell classifiers against morphologically relevant features [12].

The contribution of this dissertation is to bring these threads together on a single image dataset: a lineage-aware hierarchy derived from haematopoiesis, an imbalance-robust objective for the extreme distribution, and saliency-based verification. This differs from the common flat, single-head pipeline and is the novelty of the project.

Dataset Analysis

The selected dataset is the **MLL23 peripheral blood single-cell dataset**, released as a Data Descriptor in *Scientific Data* in 2025 [1] and hosted on Zenodo (`https://doi.org/10.5281/zenodo.14277609`). It contains **41,621** Pappenheim-stained single nucleated-cell microscopy images (288×288 pixels, $25\,\mu\text{m} \times 25\,\mu\text{m}$, TIFF), acquired at the Munich Leukemia Laboratory and annotated by expert cytomorphologists into **18** morphological classes. It is an image dataset well suited to supervised, transfer-learning, hierarchical, and imbalance-aware methods. Its defining challenge is severe class imbalance: the largest class (myeloblasts, 8,606 images) is roughly 260 times larger than the smallest (reactive lymphocytes, 33 images). For this study the 18 fine classes are grouped into three biologically motivated super-classes that form the first level of the hierarchy. Table 1 reports the per-class counts and this grouping.

Table 1: Class distribution of the MLL23 image dataset, grouped into the proposed lineage hierarchy.

Lineage (Level 1)	Cell type (Level 2)	Images
Lymphoid	Typical lymphocytes	5,532
	Hairy cells	3,265
	Large granular lymphocytes	1,849
	Atypical lymphocytes (plasma cells)	1,658
	Smudge cells	988
	Neoplastic lymphocytes	180
	Reactive lymphocytes	33
Myeloid	Myeloblasts	8,606
	Segmented neutrophils	7,170
	Monocytes	2,510
	Eosinophil granulocytes	2,448
	Atypical promyelocytes	2,033
	Myelocytes	747
	Promyelocytes	745
	Band neutrophils	687
	Basophil granulocytes	616
	Metamyelocytes	483
Erythroid	Normoblasts	2,071
Total	**18 classes**	**41,621**

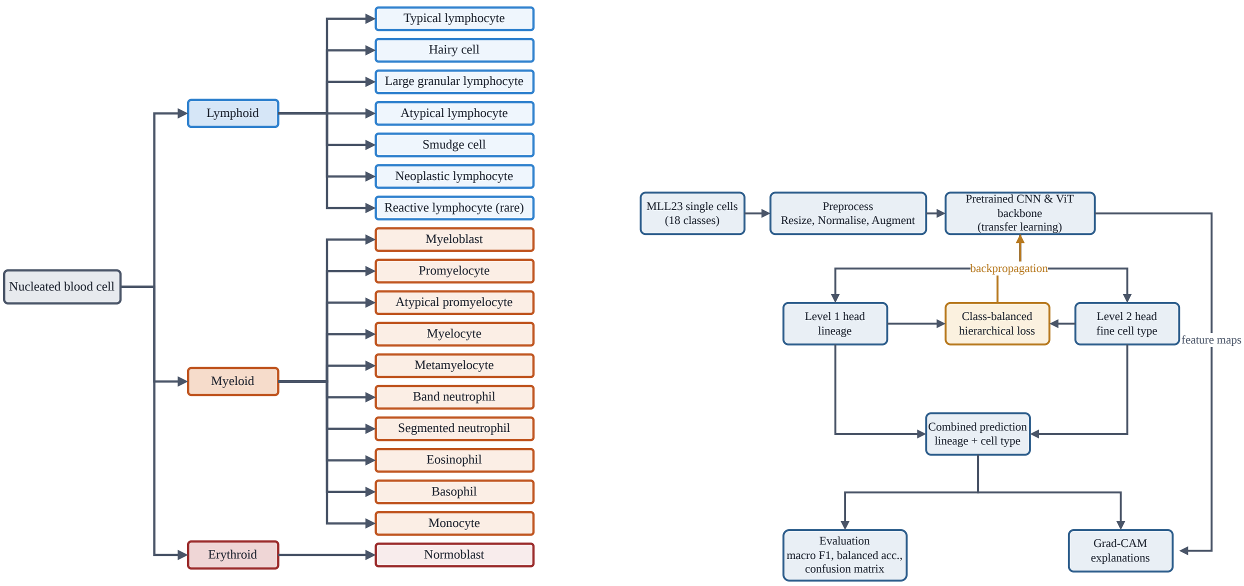
Images will be resized to 224×224 pixels to match common pretrained backbones, normalised, and split into training, validation, and test sets using a stratified scheme so every class, including the rare ones, appears in each split. Data augmentation (rotation, flipping, mild colour jitter) will be applied to the training set only, with stronger augmentation on minority classes to support generalisation.

Methodology

This research follows an experimental quantitative methodology. The central novelty is that the model is structured around blood-cell biology rather than treating the 18 labels as unrelated. Figure 1a shows the two-level lineage hierarchy (the conceptual basis of the project), and Figure 1b shows the model architecture.

Proposed model

A shared backbone feeds two heads. The Level 1 head predicts the haematopoietic lineage and the Level 2 head predicts the fine cell type. The two predictions are trained jointly, so the coarse decision regularises the fine decision and rare cell types borrow strength from their lineage neighbours. A class-balanced hierarchical



(a) Two-level label hierarchy.

(b) Overall methodology pipeline.

Figure 1: Overview of the proposed hierarchical classification framework. (a) The two-level label hierarchy, read top to bottom. Level 1 represents the haematopoietic lineage and Level 2 the fine cell type. Within the myeloid branch, cell types are ordered along the maturation continuum from immature myeloblasts to mature segmented neutrophils and other end-stage cells. (b) The overall methodology, where a shared backbone feeds both lineage and fine-type heads. During training, a class-balanced hierarchical loss updates the backbone through backpropagation, while at inference the combined prediction is evaluated and explained using Grad-CAM.

loss updates the backbone through backpropagation, while at inference the combined prediction is evaluated and explained using Grad-CAM (Figure 1b).

Processing pipeline

A structured preprocessing pipeline is applied to the raw single-cell images:

- **Resizing and normalisation:** the 288×288 TIFF images are downsampled with bilinear interpolation to 224×224 pixels, scaled to $[0, 1]$, and normalised using the ImageNet mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$) to leverage transfer-learning weights.
- **Hierarchical label mapping:** each image is mapped to a joint target $[y_1, y_2]$, where $y_1 \in \{0, 1, 2\}$ is the Level 1 lineage (Lymphoid, Myeloid, Erythroid) and $y_2 \in \{0, \dots, 17\}$ is the fine cell type from Table 1.
- **Stratified splitting:** a deterministic stratified split partitions the data into training (70%), validation (15%), and testing (15%), guaranteeing rare populations such as reactive lymphocytes appear proportionally in every set.
- **Lineage-aware augmentation:** online geometric and photometric transforms (random flips, rotation up to 90°, mild brightness/contrast jitter of 0.1) are applied during training, with an intensified regime on minority classes to enrich representation diversity.

Experimental plan

A flat single-head classifier serves as the baseline. The proposed hierarchical model is compared against it under identical data splits and backbones. An ablation removes the imbalance term, then the hierarchy in turn, to isolate each component's contribution. Because the focus is minority-class behaviour, balanced accuracy and macro F1-score are reported alongside overall accuracy, and predictions are verified qualitatively with Grad-CAM.

Tools

Table 2: Software and tools.

Tool	Purpose
Python	Main programming language
Google Colab / Jupyter	Development and experimentation with free GPU
PyTorch and Hugging Face	Backbone and hierarchical head implementation
torchvision / HF models	Pretrained backbones (ResNet50, EfficientNet-B0, MobileNetV2, ViT-Base, MobileViT)
OpenCV and Pillow	Image loading and preprocessing
NumPy and Pandas	Data handling and organisation
scikit-learn	Stratified splits and evaluation metrics
Matplotlib and seaborn	Visualisation and confusion matrices
Grad-CAM library	Saliency maps for explainability

Project Plan (Gantt Chart)

The project is scheduled across seven months, with deliberate overlap between development and evaluation so that the hierarchical model and its baseline mature together (Figure 2).

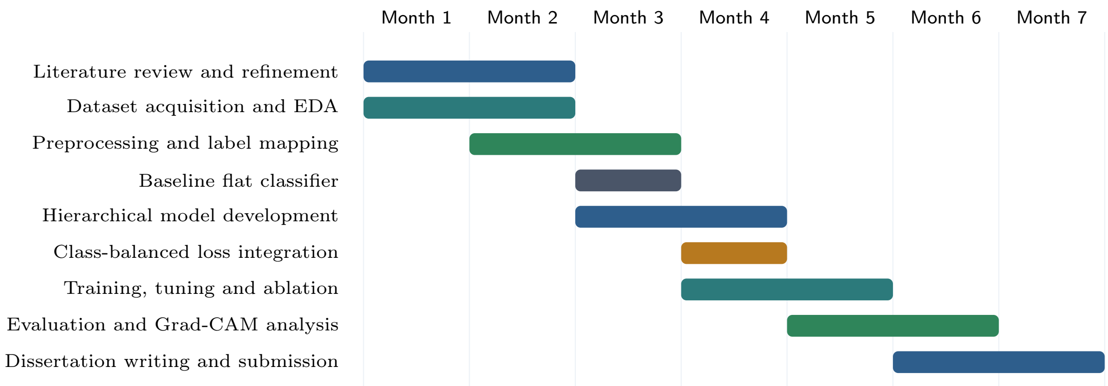


Figure 2: Indicative project timeline over seven months.

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